

Lecture 2.2 – Basic Plotting with R

Markdown

Specific Learning Objectives:

2.2.1 – Create reproducible scripts in R.

2.2.2 – Include effective documentation in scripts and projects.

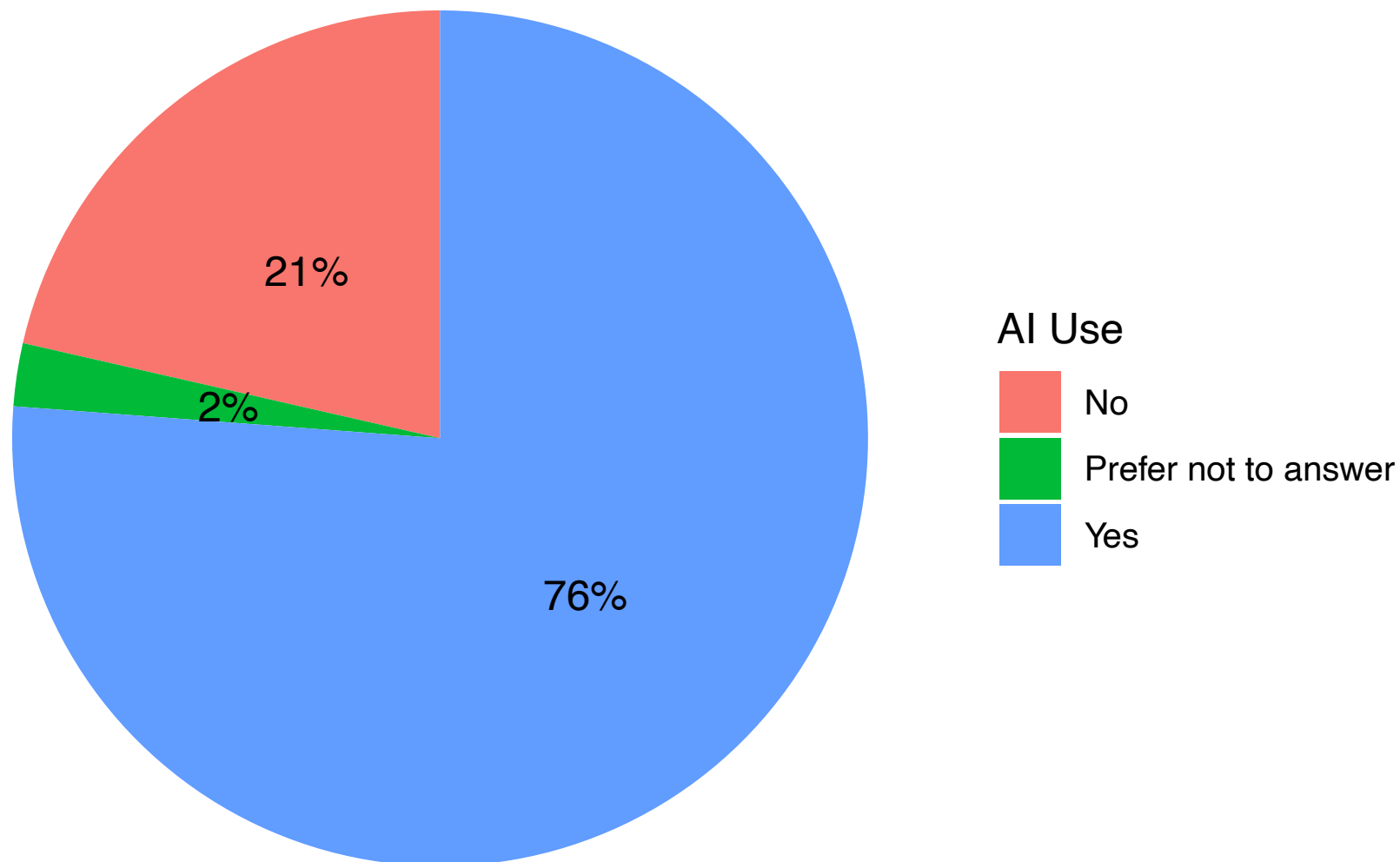
2.2.4 – Create and use Notebooks and documents using RMarkdown.

3.2 – Learn how to plot quickly using R's base graphics.

Exam 1 Debrief

- Stats presented in R Markdown Preview.
- AI use survey results

Did you use generative Artificial Intelligence when completing the outside-class assignments prior to Exam 1? (Ex: Claude, Chat GPT, etc)



Exam 1 Debrief

- AI use survey results: most common uses

- Generate practice problems

- Help with syntax or code errors

- Really stuck/time crunch/last resort for explaining an answer

- Ask for hints about where to start a problem, outline the solution to a problem.

- Help in interpreting a question

- Check answers for correctness

Exam 1 Debrief

- Help with syntax or code errors

If the first thing you do is put an error into AI, you're not going to learn how to prevent or fix that error.

Instead: read the error first, try the debugging steps we covered in lecture.

- Help in interpreting a question

If you ask for clarification before trying, this is likely unhelpful.

Instead: try the question first, put both the question and answer into AI to see if it answers the question.

- Ask for hints about where to start a problem, outline the solution to a problem.

How often are you doing this? If it is often, this is not helpful.

Instead: try anything first, take a break and come back to the problem.
Do not allow AI to take agency from you when problem solving.

Exam 1 Debrief

- Really stuck/time crunch/last resort for explaining an answer

It is better for you to leave that question blank than have AI do it.

While having AI do the question for you is perfectly allowed in the class, you are shooting yourself in the foot.

Instead: come to office hours, ask at the end of class.

Check Your Understanding

Make a plot of vapor pressure in mmHg versus temperature in C of mercury using the pressure data set. Set the axis labels and title.

Make the same plot using a different method (either with or without a formula).

Check Your Understanding

Make a plot of circumference versus age of the orange trees in the data set Orange. Give each tree a unique color *and* point shape.

Check Your Understanding

Add a legend to your graph of Orange trees.

Action Items

- 1. In class assignment L2.2**
- 2. Read Davies Chapter 8 and Chang Chapters 1-2 for next time.**